

DW_fp_sum3_DG

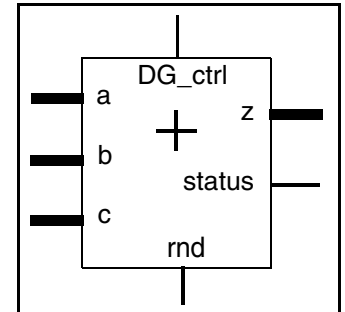
3-Input Floating-Point Adder with Datapath Gating

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Has the same functionality as DW_fp_sum3 when the component is in normal operation
- Consumes less dynamic power than DW_fp_sum3 when disabled (inputs are active, but the output is not being used)
- The precision is controlled by parameters, and covers formats in the IEEE standard
- Exponents can range from 3 to 31 bits
- Fractional part of the floating-point number can range from 2 to 253 bits
- A parameter controls the use of denormal values
- Faster than an equivalent logic using two FP adders
- Result is more accurate than using two FP adders
- DesignWare datapath generators are employed for power and QoR

Revision History



Description

DW_fp_sum3_DG is a floating-point component that adds three floating-point values, *a*, *b*, and *c*, to produce a floating-point result *z*. This component generates results with more accuracy than independent FP additions, given that only one rounding computation is done, and special conditions on the inputs are detected and corrected. Also, a control input (*DG_ctrl*) can disable the component to reduce dynamic power consumption when the component is not in use and inputs are still active.

The input *rnd* is a 3-bit rounding mode (see [Rounding Modes](#) in the *Datapath Floating-Point Overview*) and the output *status* is an 8-bit vector of status flags.

For the case of zero result, when *ieee_compliance* = 0, the sign of a zero result is negative when rounding to -infinity (*rnd* = 3), and positive for any other rounding mode. When *ieee_compliance* = 1, the following is the behavior for the sign of zero:

1. When the zero output is a result of the addition of zero inputs (all inputs are zeros), the sign of the zero output depends on the rounding mode:
 - Rounding to -infinity (*rnd* = 3): the output is +0 when all the inputs are +0, otherwise, it is -0.
 - Other rounding modes: the output is -0 when all the inputs are -0; otherwise, it is +0.
2. When the zero output is a result of the addition of non-zero inputs, the output is -0 when the rounding mode is -infinity (*rnd* = 3); otherwise, the output is +0.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	$(sig_width + exp_width + 1)$ bits	Input	Input data
b	$(sig_width + exp_width + 1)$ bits	Input	Input data
c	$(sig_width + exp_width + 1)$ bits	Input	Input data
z	$(sig_width + exp_width + 1)$ bits	Output	$(a + b) + c$
status	8 bits	Output	Status flags for the result z For details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> .
rnd	3 bits	Input	Rounding mode; supports all rounding modes described in the Datapath Floating-Point Overview
DG_ctrl	1 bit	Input	Datapath gating control 0: Component is disabled 1: Normal component operation See “ Datapath Gating Control with DG_ctrl ” on page 4

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 253 bits	Word length of fraction field of floating-point numbers a , b , c , and z
exp_width	3 to 31 bits	Word length of biased exponent of floating-point numbers a , b , c , and z
ieee_compliance	0 or 1	Level of support for IEEE 754: 0: No support for IEEE 754 NaNs and denormals; NaNs are considered infinities and denormals are considered zeros 1: Fully compliant with the IEEE 754 standard, including support for NaNs and denormals For more, see IEEE 754 Compatibility in the <i>Datapath Floating-Point Overview</i> .
arch_type	0 or 1	Controls the use of an alternative architecture. Default value is 0 (previous architecture).

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Datapath gating close to the main inputs	- DesignWare (P-2019.03 and later) - DesignWare-LP (before P-2019.03)
rtl2 ^a	Datapath gating to allow late arrival time of DG_ctrl signal	- DesignWare (P-2019.03 and later) - DesignWare-LP (before P-2019.03)

a. By default, the rtl2 implementation is used for synthesis (see “Datapath Gating Control with DG_ctrl” on page 4).

Table 1-4 Simulation Models

Model	Function
DW02.DW_FP_SUM3_DG_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_sum3_DG_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_fp_sum3_DG.v	Verilog simulation model source code

Functional Description

Table 1-5 Functional Description

a	b	c	status	z ^a
a (floating-point)	b (floating-point)	c (floating-point)	*	(a + b) + c (floating-point)

a. The actual value of the output is defined by the rounding mode.

The parameters *ieee_compliance* and *arch_type* control the functionality of this component. Different values of *arch_type* result in different numeric behaviors, but in any case, the component is always faster and more accurate than the implementation of the same function using basic floating-point components.

When the parameter *arch_type* = 0, the component provides an output that is independent of the input order (operation is totally commutative). This feature implies that the operation is done as if infinite precision was internally used, and only at the end rounding is performed to obtain the final result.

When *arch_type* = 1, the component is sensitive to the input order, in the same way that a tree of FP adders would be, when configured as (a + b) + c. Only one rounding operation is performed to compute the output value. The benefit of using this alternative is the gain in QoR for some FP formats and time constraints. The logic produced by this component when *arch_type* = 1 is usually smaller for loose time constraint than the component generated when *arch_type* = 0.

Datapath Gating Control with DG_ctrl

For DW_fp_sum3_DG and other combinational components that have the datapath gating feature, the DG_ctrl port is provided to control datapath gating.

When DG_ctrl = 1, the component behaves as expected according to activity on the input ports.

When DG_ctrl = 0:

- The component is disabled and internal gates are totally or partially isolated to block propagation of switching activity inside the component. This makes the component less sensitive to switching activity on the main ports and reduces dynamic power consumption.
- Values at the output ports are not defined.
- Simulation models set 'X' values at the output ports.

The implementations for DW_fp_sum3_DG (see [Table 1-3](#) on page 3) perform datapath gating differently:

- The rtl implementation places datapath gating as close as possible to the input ports to maximize dynamic power savings when DG_ctrl = 0. However, if the DG_ctrl signal arrives later than the data inputs, timing is degraded and area is increased to recover timing, which can increase power.
- The rtl2 implementation places datapath gating near the middle of the component. This approach is less sensitive to the arrival time of DG_ctrl and has a better chance of meeting timing and still providing dynamic power savings.

By default, the synthesis tool uses the rtl2 implementation, but you can override that. If timing constraints are loose or you know that the signal driving the DG_ctrl port arrives at the same time as other input signals, greater power savings can be attained by using the rtl implementation. You can make the override on a global level or on a case-by-case basis, as explained next.

To use the rtl implementation globally, you can disable the rtl2 implementation as follows:

- Design Compiler (before version P-2019.03):

```
set_dont_use {dw_minpower.sldb/DW_fp_div_DG/rtl2}
```
- Design Compiler (P-2019.03 and later)

```
set_dont_use {dw_foundation.sldb/DW_fp_div_DG/rtl2}
```
- Fusion Compiler:

```
set_synlib_dont_use {dw_foundation/DW_fp_div_DG/rtl2}
```

To use the rtl implementation for specific instantiated components, use the set_implementation command:

```
set_implementation U1 rtl
```

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If an invalid rounding mode has been detected on `rnd`, the following message is displayed:

```
WARNING: <instance_path>:  
        at time = <timestamp>: Illegal rounding mode.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Blocks User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_fp_sum3_DG_inst is
  generic (
    inst_sig_width : POSITIVE := 23;
    inst_exp_width : POSITIVE := 8;
    inst_ieee_compliance : INTEGER := 0;
    inst_arch_type : INTEGER := 0
  );
  port (
    inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_b : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_c : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_rnd : in std_logic_vector(2 downto 0);
    inst_DG_ctrl : in std_logic;
    z_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    status_inst : out std_logic_vector(7 downto 0)
  );
end DW_fp_sum3_DG_inst;

architecture inst of DW_fp_sum3_DG_inst is

begin

  -- Instance of DW_fp_sum3_DG
  U1 : DW_fp_sum3_DG
    generic map ( sig_width => inst_sig_width, exp_width => inst_exp_width,
    ieee_compliance => inst_ieee_compliance, arch_type => inst_arch_type )
    port map ( a => inst_a, b => inst_b, c => inst_c, rnd => inst_rnd, DG_ctrl =>
    inst_DG_ctrl, z => z_inst, status => status_inst );

end inst;
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_sum3_DG_inst( inst_a, inst_b, inst_c, inst_rnd, inst_DG_ctrl,
                          z_inst, status_inst );

parameter sig_width = 23;
parameter exp_width = 8;
parameter ieee_compliance = 0;
parameter arch_type = 0;

input [sig_width+exp_width : 0] inst_a;
input [sig_width+exp_width : 0] inst_b;
input [sig_width+exp_width : 0] inst_c;
input [2 : 0] inst_rnd;
input inst_DG_ctrl;
output [sig_width+exp_width : 0] z_inst;
output [7 : 0] status_inst;

// Instance of DW_fp_sum3_DG
DW_fp_sum3_DG #(sig_width, exp_width, ieee_compliance, arch_type)
  U1 ( .a(inst_a), .b(inst_b), .c(inst_c), .rnd(inst_rnd), .DG_ctrl(inst_DG_ctrl),
    .z(z_inst), .status(status_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Adjusted the description of the <i>ieee_compliance</i> parameter in Table 1-2 on page 2 ■ Added “ Suppressing Warning Messages During Verilog Simulation ” on page 5
March 2019	DWBB_201903.0	■ Add “ Datapath Gating Control with DG_ctrl ” on page 4 ■ Clarified some information about minPower ■ Added this Revision History table and the document links on this page

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